

Real-Time Computer Vision: Convolutional Neural Networks and Object Detection with Visual SLAM

1. Abstract & Introduction

Computer Vision applications require high-speed image processing and spatial awareness. We propose an optimized Convolutional Neural Network (CNN) architecture based on YOLO v8 for multi-class object detection and Visual Simultaneous Localization and Mapping (Visual SLAM).

2. Feature Extraction & Bounding Box Regression

The network extracts multi-scale visual features using spatial pyramid pooling. Bounding box coordinates are predicted alongside class confidence scores. Optical flow and point cloud registration enable 3D scene reconstruction from stereo camera feeds.

3. Performance Metrics

The system achieves 62 FPS at 1080p resolution with a mean Average Precision (mAP@0.5) of 88.4%. Real-time performance was verified on edge hardware devices.

4. Conclusion

Integrated feature pyramids and lightweight convolutions enable embedded computer vision systems for autonomous navigation.